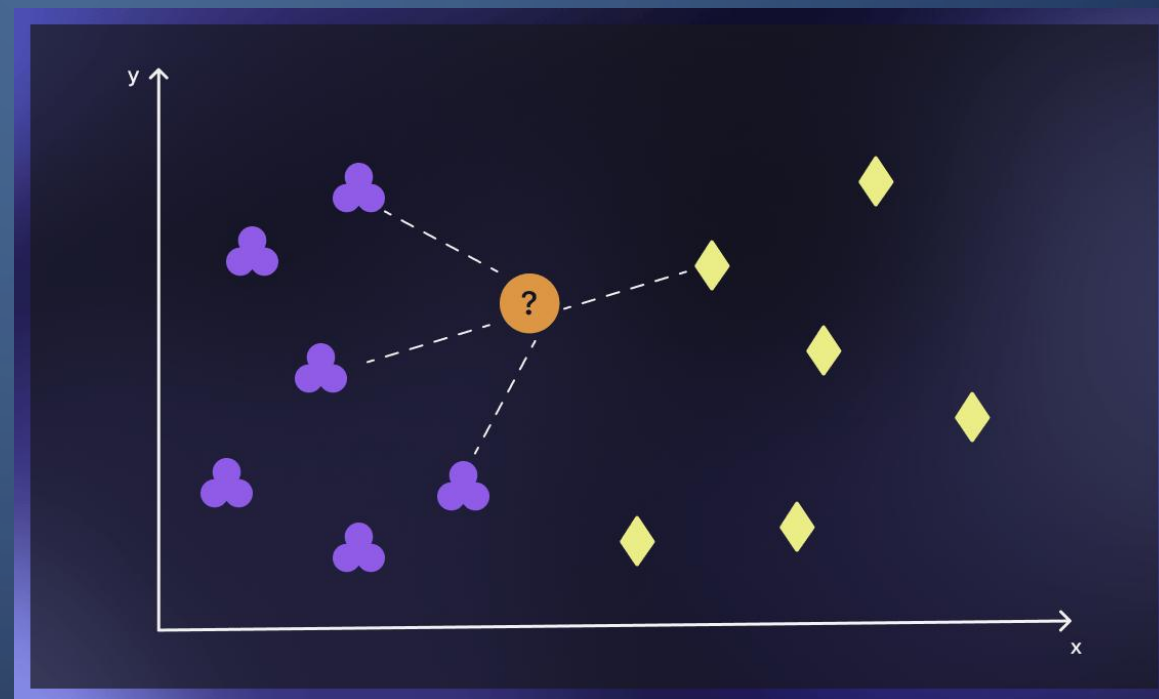


INDICATORS OF COMPROMISE USING WIRESHARK FOR DATA BREACH AT SIMONMED IMAGING BY THE RANSOMWARE GROUP MEDUSA

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INTRODUCTION

- The Medusa ransomware group is a rising threat actor in the 2025 ransomware network.
- Medusa claimed responsibility for a data breach involving SimonMed Imaging.
- The MITRE ATT&CK framework provides a structured model for analyzing adversary behavior across the cyber kill chain.
- Medusa may use enterprise tactics including Reconnaissance, Initial Access, Credential Access, Privilege Escalation, and Exfiltration.
- Use cyber tool Wireshark to find out Indicators of compromise.

RECONNAISSANCE

- The initial stage which attackers follow to gather information about their target .
- Start to collect data from publicly available information.
- Can gather all email and contact number details of the high-value targets using publicly available data.
- Reconnaissance activities are difficult to detect from the organization side.

INITIAL ACCESS

- The objective of this stage is to test if the information the team Medusa found leads to successful exploitation.
- The healthcare organizations are mostly vulnerable to phishing attacks.
- File Transfer Protocol (FTP) may also present an initial access risk.
- Remote access technology is also used to gain initial access in healthcare environments.
- A single employee mistake is enough to compromise whole system without triggering their alarm systems.

CREDENTIAL ACCESS

- Attackers try to obtain valid credentials to expand their access.
- This can be achieved through credential dumping.
- The credentials may be shared credentials or weak passwords used by the employees.
- Stolen credentials will allow attackers to authenticate the organization network as legitimate users

PRIVILEGE ESCALATION

- The team Medusa seeks to escalate privileges to administrative or system-level control.
- Privilege escalation may involve exploiting misconfigurations, abusing legitimate system tools, or leveraging compromised administrator credentials.
- Elevated privileges allows attackers to disable security controls, access sensitive servers, and prepare the environment for ransomware deployment.
- Escalation can grant attackers control over critical systems such as electronic health record servers, imaging platforms, and file shares at SimonMed Imaging.

EXFILTRATION

- Exfiltration refers to techniques adversaries use to transfer stolen data from a victim's environment to attacker-controlled systems.
- Exfiltration occurs prior to ransomware deployments, which allows Medusa team to threaten public disclosure of sensitive data if ransom demands are not met.
- These data are high value sensitive data which include protected health information (PHI), diagnostic images, billing records, insurance data, electronic health record (EHR) databases, and file server contents.
- Medusa commonly uses exfiltration over command-and-control (C2) channels.

IOC DETECTION USING WIRESHARK

- Endpoint Analysis
- Port and Connection Analysis
- FTP Session Inspection and Credential Exposure
- Evidence of File Transfer and Exfiltration

SUMMARY OF FINDINGS

- Wireshark analysis identified multiple IoC consistent with ransomware.

Severity	Finding	Evidence	Impact
High	Cleartext FTP Authentication	<i>USER jane / PASS pwd2345</i> visible in TCP stream	Immediate unauthorized access, credential leakage
High	Successful FTP Login	<i>230 Login successful</i> response	Confirms attacker gained authenticated access
High	File Transfer Activity	<i>STOR</i> command + <i>226 Transfer complete</i>	Indication data exfiltration
High	Encrypted Outbound HTTPS Traffic	Connection to 20.59.87.225 via TCP 443	Possible external relay or C2 channel
Medium	Internal Lateral Access	10.0.2.4 accessing 10.0.2.15	Lateral movement and expand
Medium	Use of Insecure Protocol (FTP)	Unencrypted file transfer	Increases likelihood of credential theft

CONCLUSION

- The data breach at SimonMed Imaging healthcare organization implies that importance of cybersecurity practices to protect from attack teams like Medusa.
- This presentation demonstrates how the group progresses through reconnaissance, initial access, credential access, privilege escalation, and exfiltration.
- Each stage contributes something to achieve their final goal of compromising the whole network and exfiltration sensitive data.
- Using cyber analytics tools like Wireshark provides valuable visibility into network activity and enables identification of indicators of compromise.
- Through network traffic analysis, suspicious behaviors such as cleartext credential exposure, unauthorized access, file transfer activity, and encrypted outbound communication can be detected before widespread damage occurs.
- SimonMed Imaging needs to have strong access control, and security awareness training to employees to mitigate these kinds of attacks.

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THANK YOU!